Over The Air Computing
Master Thesis

Department of Electrical & Computer Engineering

University of Western Macedonia

Markos Delaportas *

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Abstract – **Over-the-Air (OTA) is an emerging technique that leverages the superposition property of wireless channels to perform computations directly during the transmission process. This approach has the potential to significantly reduce latency and energy consumption in distributed systems, making it particularly suitable for applications in Internet of Things (Internet of Things (IoT)) networks, edge computing, and real-time data processing. This thesis explores the theoretical foundations of OTA, including channel modeling, signal processing techniques, and error analysis. Practical implementations and case studies demonstrating the effectiveness of OTA in various scenarios are also presented.**

Contents

*E-mail: ece01316@uowm.gr

List of Acronyms and Abbreviations

CSI	Channel State Information 4
ED	End Device 3, 5
ES	Edge Server 3
FC	Fusion Center 3, 4
IoT	Internet of Things 1
ISI	Inter-Symbol Interference 4–6
MIMO	Multiple-Input Multiple-Output 8
MSE	Mean Squared Error 4, 5
OTA	Over-the-Air 1, 3–5
PAM	Pulse Amplitude Modulation 3
SDR	Software Defined Radio 3
TDMA	Time-Division Multiple Access 7

Introduction

Over the Air Computing offers a paradigm shift in how distributed computations are performed in wireless networks. By exploiting the inherent properties of the wireless medium, OTA enables simultaneous transmissions from multiple devices to be combined in a way that directly computes a desired function of the transmitted data. This technique has garnered significant attention due to its potential to enhance efficiency and reduce latency in applications such as federated learning, distributed sensing, and real-time analytics. The motivation behind this thesis is to explore the practical implementation of OTA using Software Defined Radios Software Defined Radio (SDR) and MATLAB, addressing the challenges associated with synchronization, channel impairments, and waveform design. The research objectives include designing a prototype OTA network, validating its performance, and analyzing the impact of various impairments on computation accuracy.

0.a Background and Motivation

The increasing need for efficiency in computation-oriented applications where the interest lies in a function of local information, rather than the information itself [1]. The conventional approach of separating communication and computation often proves suboptimal [1], [2], [3]. The potential of OTA Computation is to achieve significantly higher computation rates by harnessing interference through signal superposition in a wireless Multiple Access Channel (MAC) [1].

0.b Problem Statement and Research Objectives

Focusing on designing and validating a practical OTA network layout suitable for implementation in a controllable environment[1]. The specific role of Software Defined Radios (SDRs) and MATLAB/simulation tools for prototype validation and addressing practical limitations such as synchronization and channel impairments [1], [4], [5].

0.c Thesis Contributions and Organization

The aim of this thesis is to contribute to the research objectives...as a student member of the university of western macedonia, department of electrical and computer engineering, under the supervision of professor ....

1 System Model and Practical Challenges in OTA

A system that utilizes OTA computing can be modeled in various ways depending on the specific use case and propagation environment. Most architectures involve a set of End Device (ED)s that transmit concurrently to a Fusion Center (FC) or Edge Server (ES). The communication channel is a critical component of this model, as it introduces physical impairments such as fading, noise, and path loss. In addition to the channel model, practical challenges inherent to wireless communications—such as time sampling errors, multipath propagation, and waveform design—can significantly degrade the computational reliability of OTA systems. To mitigate these effects, optimization strategies including power management, transceiver design, and equalization must be employed.

1.a System Architecture and Channel Model

To mathematically formulate the impact of channel impairments on OTA computing, we consider a discrete-time network model consisting of multiple EDs deployed in an agricultural environment. These devices utilize Pulse Amplitude Modulation (PAM) to transmit local temperature readings to a central FC.

Let $x_i[n]$ denote the discrete-time signal transmitted by the i-th sensor. For an illustrative scenario involving two sensors, the transmitted sequences are defined as:

$$x_1[n] = [2, -1, 1] \tag{1}$$
$$x_2[n] = [1, 2, -1] \tag{2}$$

1.a.i The Ideal Target Function

The primary objective of OTA computing is to leverage the multiple access channel (MAC) superposition property to aggregate data natively in the air. Assuming a flat-fading, noise-free, and perfectly synchronized channel, the ideal target function received at the FC, denoted as $s[n]$, is the direct sum of the transmitted signals:

$$s[n] = \sum_{i=1}^{K} x_i[n] \tag{3}$$

where K is the total number of transmitting devices. For the given sensor data, the ideal aggregated sequence is evaluated as $s[n] = [3, 1, 0]$.

1.a.ii Multipath Channel Distortion

In practical agricultural deployments, the physical environment exhibits multipath characteristics (e.g., reflections from terrain or machinery). This is modeled as a Finite Impulse Response (FIR) filter. Assuming a channel impulse response $h[n]$ with a span of three taps, defined as $h[n] = [1.0, 0.5, -0.2]$, the actual received signal $y[n]$ is the convolution of the ideal aggregated signal with the channel response:

$$y[n] = s[n] * h[n] = \sum_{k=0}^{n} s[k]h[n-k] \tag{4}$$

Evaluating this convolution yields the distorted sequence $y[n] = [3.0, 2.5, -0.1, -0.2, 0.0]$. The extension of the sequence length from three to five symbols mathematically illustrates the presence of Inter-Symbol Interference (ISI), wherein previous and future symbols bleed into the current sampling instance.

1.a.iii Mean Squared Error (MSE) Evaluation

To quantify the degradation caused by the multipath channel, the Mean Squared Error (MSE) is calculated between the ideal target function $s[n]$ and the actual received signal $y[n]$. The error at each discrete time step is given by $e[n] = y[n] - s[n]$. For a sequence of length N, the overall MSE is defined as:

$$\text{MSE} = \frac{1}{N} \sum_{n=0}^{N-1} (y[n] - s[n])^2 \tag{5}$$

For the evaluated sequences, the discrepancy between the ideal sum and the ISI-corrupted signal results in an MSE $= 0.46$. This theoretical error establishes the baseline challenge that must be addressed through waveform design or digital equalization techniques at the FC.

1.b Practical Challenges

1.b.i Channel State Information (CSI) Imperfection

The theoretical model assumes perfect knowledge of the channel impulse response. In practice, the need for accurate and fresh Channel State Information (CSI) at transmitters or receivers is critical for precoding and equalization techniques [1],

[5], [6]. Obsolete or noisy CSI severely limits the ability to pre-distort signals, exacerbating the MSE.

1.b.ii Waveform and Filter Design

As demonstrated by the sequence expansion in the multipath model, standard digital communication transceivers suffer from time sampling errors and ISI. Designing custom pulse shapes, such as optimized Root Raised Cosine (RRC) filters or DNN-generated waveforms, is essential to force zero-crossings at the exact sampling instances, thereby suppressing ISI natively over the air [5].

1.c Optimization Strategies to Enhance OAC Performance

To mitigate the aforementioned challenges, several optimization frameworks have been proposed in the literature:

- **Power Management**: Strategies to mitigate aggregation errors involve optimizing power control policies at the EDs to align received signal amplitudes, thereby minimizing the overall MSE or maximizing the convergence speed of federated learning algorithms [1], [5], [7].
- **Transceiver Design**: The joint design of pre-processing functions at the transmitter and post-processing equalizers at the receiver allows the system to match the channel characteristics to the desired nomographic function, effectively canceling multipath echoes and minimizing computational distortion [1], [3], [6], [8].

2 Experimental Methodology and Implementation

2.a Analog OAC and Nomographic Functions

To validate the mathematical models established in the previous section, an experimental framework was developed. The implementation focuses on analog OTA to compute nomographic functions—specifically, the arithmetic mean of distributed temperature sensors. By leveraging the superposition property of the multiple access channel, the network natively aggregates uncoded sensor readings in the air, transforming the physical medium into a computational processor.

2.b GNU Radio System Architecture

To physically simulate the analog OTA network, a Software Defined Radio (SDR) architecture was constructed using GNU Radio. The simulation environment accurately models the concurrent transmission of multiple Edge Devices (EDs) to a central Fusion Center (FC).

2.b.i Hardware and Software Parameters

The baseline configuration for the SDR implementation is detailed in Table 2. These parameters were selected to balance computational load while providing sufficient bandwidth to observe channel impairments.

2.b.ii Pulse Shaping and Channel Impairments

Two critical components of the GNU Radio flowgraph govern the physical behavior of the transmission:

- **Root Raised Cosine (RRC) Filter**: To constrain the transmission bandwidth and mitigate Intersymbol Interference (ISI), RRC filters are employed for pulse shaping at both the transmitters and the receiver. The RRC filter ensures that, under ideal Nyquist sampling rates, the overlapping tails of adjacent symbols evaluate to zero, thereby preserving the integrity of the target function computation.

Table 2: Key SDR Hardware and Software Parameters for the OAC Implementation

Parameter	Value	Description
Hardware Configurations		
Center Frequency (f_c)	2.4 GHz	ISM band, selected to minimize external interference.
Sample Rate (f_s)	32 kHz	Defined baseband rate for the SDR sink/source blocks.
Antenna Configuration	SISO	Single-Input Single-Output for initial baseline testing.
Software & Baseband Processing		
Modulation Scheme	PAM	Pulse Amplitude Modulation for analog temperature transmission.
Samples per Symbol (sps)	6	Determines the interpolation/decimation rate for the signal.
Pulse Shaping Filter	RRC	Root Raised Cosine filter applied to mitigate ISI.
Roll-off Factor (α)	0.3	Standard excess bandwidth factor defining the filter's frequency transition band.

- **Channel Model**: To accurately emulate an agricultural deployment, a Channel Model block is introduced to inject realistic wireless impairments. This block models Additive White Gaussian Noise (AWGN) and user-defined multipath delay taps, which scatter the signal arrivals and intentionally induce ISI to test the system's robustness.

2.b.iii Flowgraph Configurations

The network was simulated under two primary conditions. Figure 1 illustrates the continuous-time analog simulation where sensor data is continuously evaluated. Figure 2 depicts the "Faster-Than-Nyquist" (FTN) scenario, wherein the transmission rate is artificially accelerated to break the Nyquist limit and observe the resulting ISI degradation.

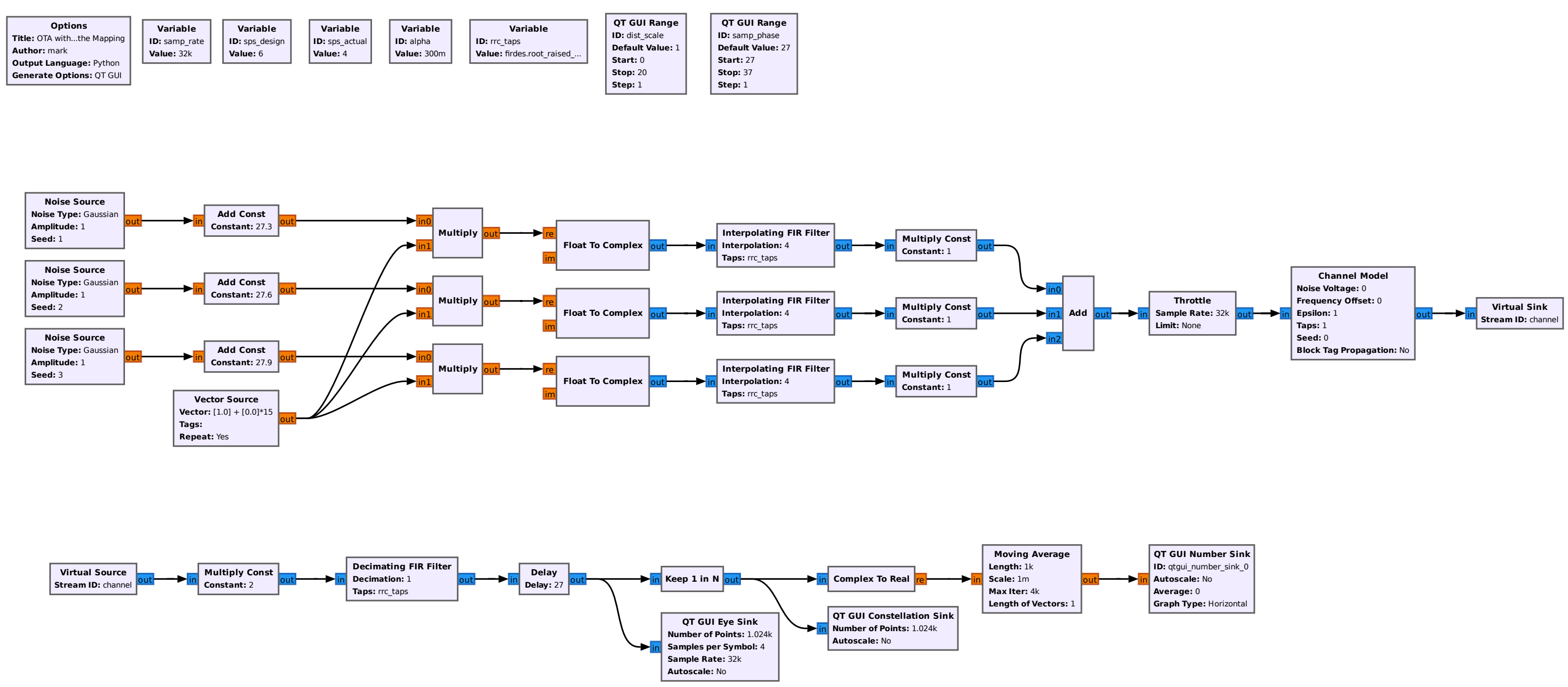

Figure 1: GNU Radio Companion flowgraph for the OTA implementation including Channel Model and RRC filters.

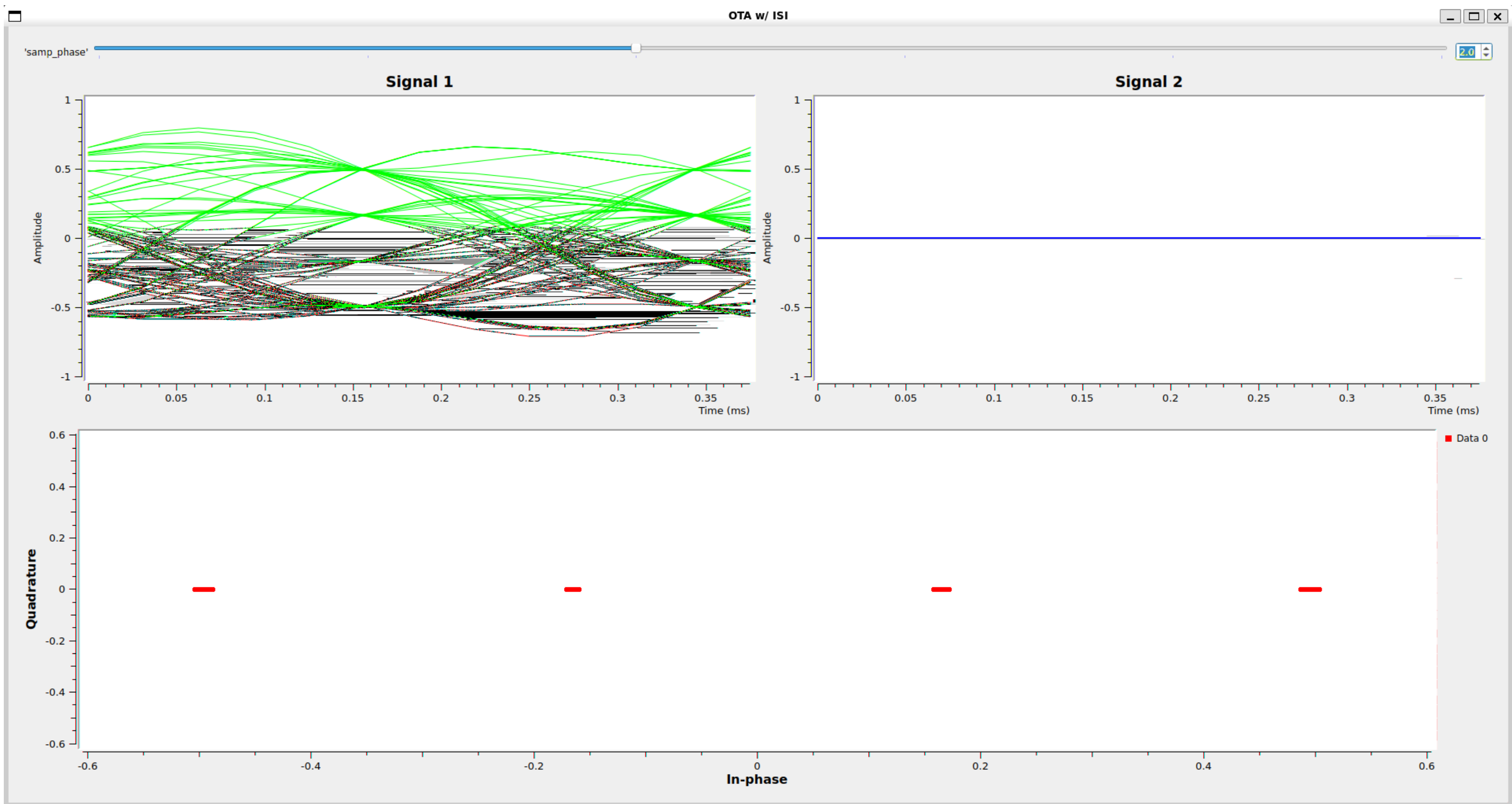

Figure 2: GNU Radio Companion flowgraph configured to induce Intersymbol Interference (ISI) by mismatching the design and actual sampling rates.

2.c MATLAB and Simulink Integration

While GNU Radio provides a robust framework for real-time SDR execution and analog signal evaluation, future phases of this methodology will incorporate MATLAB Simulink. Simulink will be utilized to model advanced feedback loops, custom waveform generation, and digital equalizers (e.g., Minimum Mean Square Error (MMSE) receivers) that are mathematically complex to implement in standard GNU Radio environments. This dual-software approach allows for both practical SDR validation and rigorous mathematical algorithmic design.

3 Experimental Results and Discussion

3.a Validation of the Implemented OAC Network

Demonstration of successful signal superposition and computation via the SDR platform[1], [2]. Comparison of achieved computation result against the ideal theoretical value[1], [9].

3.b Performance Evaluation under Impairments

Quantifying the performance metrics, such as Mean Squared Error (MSE) or Computation Error Rate (CER), of the implemented scheme[1], [5]. Analysis of the impact of deliberately introduced synchronization errors or varying channel noise (SNR) on the computation accuracy[1], [5]. Comparison with Separation Schemes (Simulated Baseline); Illustrating the resource savings and computation rate achieved by the OAC approach relative to traditional orthogonal access (e.g., Time-Division Multiple Access (TDMA))[1], [2], [3].

4 Conclusion and Future Work

4.a Summary of Thesis Findings

Recap of the feasibility and performance demonstrated by the SDR-based OAC network.

4.b Future Research Directions

Exploring more complex OAC architectures, such as multi-antenna systems Multiple-Input Multiple-Output (MIMO) or using Reconfigurable Intelligent Surfaces (RIS)[5], [10]. Further investigation into standardized OAC protocols for compatibility with modern wireless networks (e.g., LTE/5G protocols)[1], [9].

A MATLAB Code for OTA Simulation

This appendix contains the core MATLAB scripts used to generate the simulation data discussed in Chapter 2. Note that we utilized the Raised Cosine method [1].

References

[1] A. Şahin and R. Yang, "A survey on over-the-air computation," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 3, pp. 1877–1908, 2023, ISSN: 1553-877X, 2373-745X. DOI: 10.1109/COMST.2023.3264649 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10092857/

[2] M. Goldenbaum, H. Boche, and S. Stanczak, "Analyzing the space of functions analog-computable via wireless multiple-access channels," in *2011 8th International Symposium on Wireless Communication Systems*, Aachen, Germany: IEEE, Nov. 2011, pp. 779–783, ISBN: 978-1-61284-402-2 978-1-61284-403-9 978-1-61284-401-5. DOI: 10.1109/ISWCS.2011.6125268 Accessed: Dec. 4, 2025. [Online]. Available: http://ieeexplore.ieee.org/document/6125268/

[3] M. Goldenbaum and S. Stanczak, "Robust analog function computation via wireless multiple-access channels," *IEEE Transactions on Communications*, vol. 61, no. 9, pp. 3863–3877, Sep. 2013, ISSN: 0090-6778. DOI: 10.1109/TCOMM.2013.072913.120815 Accessed: Dec. 4, 2025. [Online]. Available: http://ieeexplore.ieee.org/document/6573232/

[4] Y. Xiao, Y. Ye, S. A. Tegos, P. D. Diamantoulakis, Y. You, and G. K. Karagiannidis, "Heterogeneous wireless federated learning framework via over-the-air computation," *IEEE Transactions on Vehicular Technology*, pp. 1–14, 2025, ISSN: 0018-9545, 1939-9359. DOI: 10.1109/TVT.2025.3600821 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/11131470/

[5] N. G. Evgenidis et al., "Waveform design for over-the-air computing," *IEEE Transactions on Wireless Communications*, pp. 1–1, 2025, ISSN: 1536-1276, 1558-2248. DOI: 10.1109/TWC.2025.3613838 arXiv: 2405.20877[cs]. Accessed: Dec. 4, 2025. [Online]. Available: http://arxiv.org/abs/2405.20877

[6] K. Yang, T. Jiang, Y. Shi, and Z. Ding, "Federated learning via over-the-air computation," *IEEE Transactions on Wireless Communications*, vol. 19, no. 3, pp. 2022–2035, Mar. 2020, ISSN: 1536-1276, 1558-2248. DOI: 10.1109/TWC.2019.2961673 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/8952884/

[7] X. Cao, G. Zhu, J. Xu, Z. Wang, and S. Cui, "Optimized power control design for over-the-air federated edge learning," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 1, pp. 342–358, Jan. 2022, ISSN: 0733-8716, 1558-0008. DOI: 10.1109/JSAC.2021.3126060 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9606731/

[8] H. Hellström, S. Razavikia, V. Fodor, and C. Fischione, "Optimal receive filter design for misaligned over-the-air computation," in *2023 IEEE Globecom Workshops (GC Wkshps)*, Kuala Lumpur, Malaysia: IEEE, Dec. 4, 2023, pp. 1529–1535, ISBN: 979-8-3503-7021-8. DOI: 10.1109/GCWkshps58843.2023.10464656 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10464656/

[9] G. Zhu, Y. Du, D. Gunduz, and K. Huang, “One-bit over-the-air aggregation for communication-efficient federated edge learning: Design and convergence analysis,” *IEEE Transactions on Wireless Communications*, vol. 20, no. 3, pp. 2120–2135, Mar. 2021, ISSN: 1536-1276, 1558-2248. DOI: 10.1109/TWC.2020.3039309 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9272666/

[10] J. Zhu, Y. Shi, Y. Zhou, C. Jiang, W. Chen, and K. B. Letaief, “Over-the-air federated learning and optimization,” *IEEE Internet of Things Journal*, vol. 11, no. 10, pp. 16 996–17 020, May 15, 2024, ISSN: 2327-4662, 2372-2541. DOI: 10.1109/JIOT.2024.3352280 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10388035/